

Bishal Shrestha

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RESEARCH INTERESTS

Artificial intelligence and computational biology, deep learning models for genomics and 3D chromatin organization, single-cell genomics, graph neural networks and transformers for biological data, protein structure prediction and function annotation, spatiotemporal analysis of Hi-C data, and developing computational tools for understanding genome architecture and its role in human health and disease.

RESEARCH EXPERIENCE

Research Assistant

Aug 2023 – Present

University of Miami, Coral Gables, FL

- Developed HiC4D-SPOT, an unsupervised deep-learning framework using ConvLSTM-based autoencoders to detect spatiotemporal anomalies in Hi-C data, published in *Briefings in Bioinformatics* (2025); achieved high reconstruction fidelity (PCC/SCC > 0.9) and accurately identified structural anomalies including TAD disruptions and chromatin remodeling events.
- Developed scHiGex, a Graph Transformer-based model for predicting single-cell gene expression using simultaneously captured single-cell Hi-C data, published in *NAR Genomics and Bioinformatics* (2025); achieved average absolute error of 0.07 and precise cell type classification (adjusted Rand index = 1).
- Designed and implemented an ensemble of graph neural network architectures to estimate local residue accuracy in protein structure models as part of the CASP 16, a leading competition in computational biology.
- Implemented Geometric Algebra Transformers to annotate protein functions for protein models, showcasing use of geometric deep learning in structural biology.

Research Intern

Sep 2021 – Aug 2022

University of Missouri - St. Louis, MO

- Conducted research on adversarial sample generation for Nepal's embossed license plate recognition system, employing adversarial training to enhance model robustness against attack scenarios.
- Designed and executed deep learning experiments using convolutional neural networks (CNNs) in PyTorch, including collecting and preprocessing a novel dataset for real-world applications.

Student Researcher

Apr 2021 – Sep 2021

Research and Innovation Unit, Tribhuvan University

- Investigated machine learning techniques for customer churn prediction, collaborating with a leading Nepali ISP to optimize business operations through predictive analytics.
- Presented findings at the Himalaya International Conference, highlighting the impact of academia-industry partnerships in addressing real-world problems.

EDUCATION

PhD in Computer Science

Aug 2023 - Present

University of Miami, Coral Gables, FL

Advisor: Dr. Zheng Wang

Bachelor's in Computer Engineering

Jun 2017 - Jan 2022

Tribhuvan University, Nepal

Advisor: Dr. Badri Adhikari

TEACHING

Visiting Lecturer

Spring 2025, Fall 2024

University of Miami, Coral Gables, FL

- CSC314: Computer Organization and Architecture:** Spring 2025, Fall 2024 (2 sections)

Graduate Teaching Assistant

Spring 2025, Spring 2024, Fall 2023

University of Miami, Coral Gables, FL

- CSC402: Computer Science Practicum II:** Spring 2025, Spring 2024
Student Course Evaluations: Achieved **4.9/5** (Spring 2024) and **4.8/5** (Spring 2025)
- CSC113/CSC200: Data Science for the World:** Fall 2023

PROGRAMMING SKILLS

Programming Languages: Python, C, C++

Machine Learning Frameworks: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas

Data Visualization: Matplotlib, Seaborn

Certifications: Mathematics for Machine Learning (Imperial College London), Neural Networks and Deep Learning (Coursera), Qiskit Global Summer School (IBM)

Soft Skills: Effective communication, teamwork, problem-solving, critical thinking

REFERENCES

- **Dr. Zheng Wang:** Associate Professor, Department of Computer Science, University of Miami
Email: zheng.wang@miami.edu; Phone: +1 (305) 284-3642
Address: 330M Ungar Building, 1365 Memorial Drive, Coral Gables, FL 33124-4245
(PhD Advisor)
- **Dr. Badri Adhikari:** Associate Professor, Department of Computer Science, University of Missouri - St. Louis
Email: adhikarib@umsl.edu; Phone: +1 (573) 639-2847
Address: Department of Computer Science, University of Missouri - St. Louis, 1 University Boulevard, St. Louis, MO 63121
(Research Mentor)

PUBLICATIONS

1. **Shrestha B**, Wang Z. “*HiC4D-SPOT: a spatiotemporal outlier detection tool for Hi-C data.*” Briefings in Bioinformatics, 26(4), bbaf341. 2025.
2. **Shrestha B**, Siciliano AJ, Zhu H, Liu T, Wang Z. “*scHiGex: predicting single-cell gene expression based on single-cell Hi-C data.*” NAR Genomics and Bioinformatics. 7 (1), Iqaf002. 2025.
3. **Shrestha B**, Khakurel G, Simkhada K, Adhikari B. “*Adversarial sample generation and training using geometric masks for accurate and resilient license plate character recognition.*” arXiv preprint arXiv:2311.12857. 2023.
4. Prajapati S, Engelhardt M, Shrestha A, Khakurel G, **Shrestha B**, Simkhada K. “*Industry-Academia Collaboration in Nepal - A case study of Customer Churn Prediction for Worldlink.*” Research and Innovation Unit. 2021.

PUBLICATIONS UNDER REVIEW

5. Zhu H, Liu T, **Shrestha B**, Wang Z. “*SCW: building the whole-genome 3D structures based on extremely sparse single-cell Hi-C data*” BMC Bioinformatics, Submitted Dec 2025.
6. Di Giovanni S, Mueller F, Maclachlan E, Costa A, Qu J, **Shrestha B**, Wang Z, De Virgiliis F, Hutson T, Zhou L, Kong G, Chadwick J, Sousa M, Palmisano I. “*Druggable CITED2 Is a Reversible Epigenetic Switch in Neuronal Maturation to Regenerative Decline.*” EMBO Molecular Medicine, Resubmitted Dec 2025.
7. Siciliano AJ, Bao Y, **Shrestha B**, Wang Z. “*Inferring the qualities of protein-RNA models with graph transformers.*” Bioinformatics, Submitted Nov 2025.

PROFESSIONAL SERVICE

Reviewer

2026

ISMB (Intelligent Systems for Molecular Biology)

POSTER AND PRESENTATIONS

1. **Shrestha B** (Advisor: Z. Wang) “*HiC4D-SPOT: a spatiotemporal outlier detection tool for Hi-C data*” Research Presentation at the annual Computing Day, University of Miami, Coral Gables, FL, April 2025.
2. **Shrestha B** (Advisor: Z. Wang) “*Predicting single-cell gene expression based on single-cell Hi-C using graph transformer*” Research Presentation at the annual Computing Day, University of Miami, Coral Gables, FL, April 2024.
3. **Shrestha B** (Advisor: Z. Wang) “*Predicting single-cell gene expression based on single-cell Hi-C using graph transformer*” Research Presentation at the Sylvester Annual Retreat, Sylvester Comprehensive Cancer Center, University of Miami, Miami, FL, Oct 2024.